Solving Exponential Equations by Writing a Common Base

Equating exponents when the bases already match, rewriting both sides over a common base, and solving exponential models the same way

The rule and the model. For $b > 0$ and $b \neq 1$, the function b^u is one-to-one, so

$$b^u = b^v \iff u = v.$$

Matching bases is therefore not a trick — it is what licenses dropping the bases and solving the ordinary equation left behind. Use this table of small powers to spot a shared base:

Power	b^1	b^2	b^3	b^4	b^5	b^6	b^7	b^8
Base 2	2	4	8	16	32	64	128	256
Base 3	3	9	27	81	243	—	—	—
Base 5	5	25	125	625	—	—	—	—

- Read every number in the equation off the table. If they are all powers of one base, rewrite each side as that base raised to a power, using $(b^m)^n = b^{mn}$ and $b^m b^n = b^{m+n}$.
- A reciprocal is a negative exponent: $\frac{1}{2} = 2^{-1}$ and $\frac{1}{125} = 5^{-3}$.
- Once both sides read $b^{\text{something}}$, set the two exponents equal and solve. Then substitute back to check.
- Show the rewriting line before the exponent equation — that line is where the work is.

1. The bases already match. Solve for x.

$$2^{x+1} = 2^{3x-5}$$

Answer: ________________

2. The bases already match. Solve for x.

$$3^{2x} = 3^{x+4}$$

Answer: ________________

3. Write the right-hand side as a power of 3 first, then solve for x. Leave the answer as a fraction.

$$9^{x} = 27$$

Answer: ________________

4. Solve for x, then check your answer by substituting it back into both exponents.

$$5^{3x-1} = 5^{x+9}$$

Answer: ________________

5. Both sides are powers of 2. Rewrite each side over base 2, then solve for x.

$$4^{2x} = 8^{x+1}$$

Answer: ________________

6. Write both sides as powers of 2 — remember that $\frac{1}{2} = 2^{-1}$ — then solve for x.

$$\left(\tfrac{1}{2}\right)^x = 32$$

Answer: ________________

7. Doubling model. A culture starts with 5 cells and doubles every hour, so after t hours the count is $N = 5 \cdot 2^t$.

Hours t	0	1	2	3	4
Cells N	5	10	20	40	80

Extend the pattern in your head, then solve $5 \cdot 2^t = 320$ algebraically: divide by 5 first so that the equation becomes a single power of 2 on each side. How many hours does it take?

Answer: ________________

8. Check the claim. A student says that $x = -2$ solves

$$25^{x-1} = 125^x.$$

Rewrite both sides over base 5 and solve the equation yourself. Then confirm the claim by computing the difference $25^{x-1} - 125^x$ at $x = -2$, and state what value that difference must have if the claim is true.

Answer: ________________

9. Decay model. A 240 mg sample loses half its mass every 5 years, so the mass remaining after t years is $M = 240\left(\frac{1}{2}\right)^{t/5}$. Solve

$$240\left(\tfrac{1}{2}\right)^{t/5} = 15$$

for t by writing both sides as powers of 2. Say how many half-lives that is, and explain why the answer had to be a whole number of half-lives here.

Answer: ________________

10. Challenge. First write $9^x \cdot 27^{x+1}$ as a single power of 3. Then use that form to solve

$$9^x \cdot 27^{x+1} = 81.$$

Give both the single-power form and the value of x.

Answer: ________________